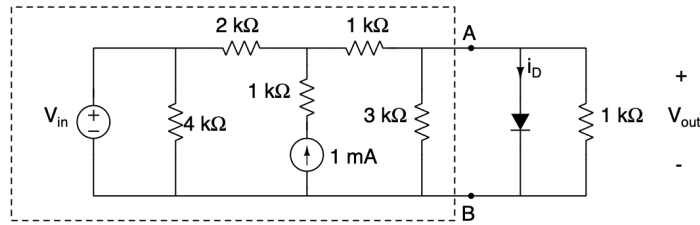


Question 1 [25 Points]:



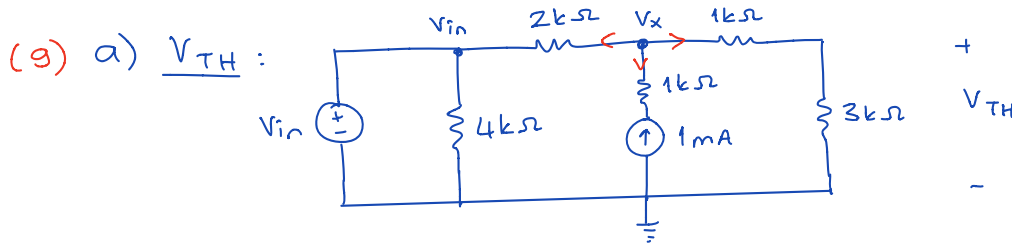
For the circuit given above, assume that $V_{ON} = 0.6\text{ V}$ for the diode.

(a) Find the Thevenin equivalent of the circuit inside the dashed box, between the terminals A and B. Express the Thevenin equivalent voltage in terms of V_{in} .

(b) Find V_{out} and i_D if $V_{in} = 2\text{ V}$.

(c) Find V_{out} and i_D if $V_{in} = -4\text{ V}$.

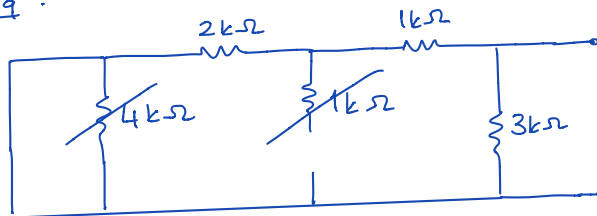
Hint: For parts (b) and (c), use your results from part (a).



$$\text{KCL at } V_x: \frac{V_x - V_{in}}{2k} - 1\text{mA} + \frac{V_x}{4k} = 0 \Rightarrow V_x = \frac{2V_{in} + 4}{3}$$

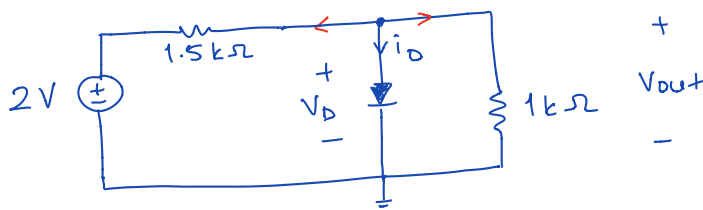
$$V_{TH} = V_x \cdot \frac{3k}{1k + 3k} = \frac{2V_{in} + 4}{4} = \boxed{\frac{V_{in} + 2}{2}} \quad (\text{voltage divider})$$

R_{eq} : Kill all sources



$$R_{eq} = (2k + 1k) // 3k = \boxed{1.5k\Omega}$$

(8) b) $V_{in} = 2\text{ V} \Rightarrow V_{TH} = \frac{2+2}{2} = 2\text{ V} \Rightarrow$ Assume diode is ON

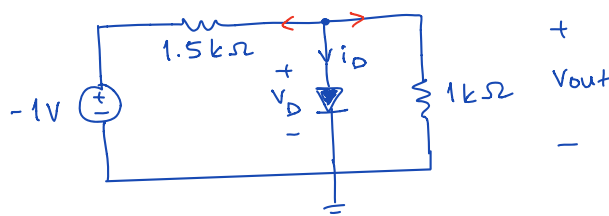


$$V_D = V_{out} = \boxed{0.6\text{ V}}$$

$$\text{KCL: } i_D + \frac{0.6 - 2}{1.5k} + \frac{0.6}{1k} = 0$$

$$i_D = \boxed{\frac{1}{3}\text{ mA}} > 0 \quad (\text{confirms diode is ON})$$

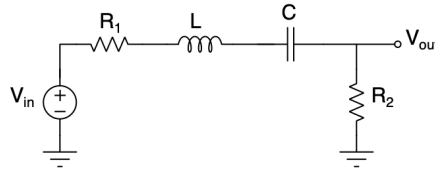
(8) c) $V_{in} = -4\text{ V} \Rightarrow V_{TH} = \frac{-4+2}{2} = -1\text{ V} \Rightarrow$ Assume diode is OFF



$$\boxed{i_D = 0}$$

$$V_{out} = -1 \cdot \frac{1k}{1.5k + 1k} = \boxed{-0.4\text{ V}}$$

$$V_D = V_{out} = -0.4\text{ V} < 0 \quad (\text{confirms diode is OFF})$$

Question 2 [30 Points]:

For the circuit given above, $R_1 = 20 \Omega$, $R_2 = 80 \Omega$, $L = 2 \mu H$, $C = 50 pF$.

- (a) Find the equivalent impedance $Z_{eq}(\omega)$ seen by the voltage source. Express $Z_{eq}(\omega)$ in parametric form in terms of R_1 , R_2 , L , C (i.e., do not insert component values).
- (b) Using the component values, find the resonance frequency ω_0 , quality factor Q , 3-dB bandwidth $\Delta\omega$ for the tuned circuit. Also, find the 3-dB cut-off frequencies ω_1 and ω_2 , where $\omega_2 > \omega_1$.
- (c) Based on your answers in (a) and (b), roughly plot $|Z_{eq}(\omega)|$. Mark important points and values in your plot.
- (d) Find $v_{out}(t)$ if $v_{in}(t) = 2 \cos(\omega_0 t)$ V.
- (e) Find $v_{out}(t)$ if $v_{in}(t) = 2 \cos(\omega_1 t)$ V.

(3) a) $Z_{eq}(\omega) = R_1 + R_2 + j\omega L + \frac{1}{j\omega C} = (R_1 + R_2) + j\left(\frac{\omega^2 LC - 1}{\omega C}\right)$

(10) b) $\text{Im}\{Z_{eq}(\omega)\} = 0 \Rightarrow \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{2 \cdot 10^{-6} \cdot 50 \cdot 10^{-12}}} = 10^8 \text{ rps}$

$$Q = \frac{\omega_0 L}{R_1 + R_2} = \frac{10^8 \cdot 2 \cdot 10^{-6}}{100} = 2$$

$$Q = \frac{\omega_0}{\Delta\omega} \Rightarrow \Delta\omega = \frac{\omega_0}{Q} = \frac{10^8}{2} = 5 \cdot 10^7 \text{ rps}$$

$$\omega_0^2 = \omega_1 \cdot \omega_2 = 10^{16} \Rightarrow \omega_2 = \frac{10^{16}}{\omega_1}$$

$$\Delta\omega = \omega_2 - \omega_1 = 5 \cdot 10^7 \Rightarrow \frac{10^{16}}{\omega_1} - \omega_1 = 5 \cdot 10^7$$

$$\omega_1^2 + 5 \cdot 10^7 \omega_1 - 10^{16} = 0$$

$$\omega_1 = \frac{-5 \cdot 10^7 \pm \sqrt{25 \cdot 10^{14} + 4 \cdot 10^{16}}}{2} = \frac{10^7}{2} (-5 + \sqrt{425}) = 7.81 \times 10^7 \text{ rps}$$

$$\omega_2 = \Delta\omega + \omega_1 = 12.81 \times 10^7 \text{ rps}$$

(6) c) $|Z_{eq}(\omega)| = \sqrt{(R_1 + R_2)^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$

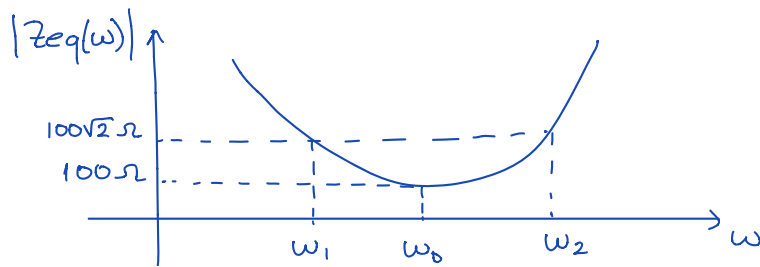
* As $\omega \rightarrow 0$, $|Z_{eq}(\omega)| \rightarrow \infty$

* As $\omega \rightarrow \infty$, $|Z_{eq}(\omega)| \rightarrow \infty$

* At $\omega = \omega_0$, $|Z_{eq}(\omega_0)| = R_1 + R_2 = 100 \Omega$

$$\begin{aligned}
 * \text{ At } \omega = \omega_1, \quad Z_{eq}(\omega_1) &= R_1 + R_2 + j\omega_1 L + \frac{1}{j\omega_1 C} \\
 &= 100 + j7.81 \times 10^7 \times 2 \times 10^{-6} + \frac{1}{j7.81 \times 10^7 \times 50 \times 10^{-12}} \\
 &= 100 - j100 \, \Omega \\
 |Z_{eq}(\omega_1)| &= 100\sqrt{2} \, \Omega \quad (\text{as expected})
 \end{aligned}$$

$$\begin{aligned}
 * \text{ At } \omega = \omega_2, \quad Z_{eq}(\omega_2) &= R_1 + R_2 + j\omega_2 L + \frac{1}{j\omega_2 C} \\
 &= 100 + 12.81 \times 10^7 \times 2 \times 10^{-6} + \frac{1}{j12.81 \times 10^7 \times 50 \times 10^{-12}} \\
 &= 100 + j100 \, \Omega \\
 |Z_{eq}(\omega_2)| &= 100\sqrt{2} \, \Omega \quad (\text{as expected})
 \end{aligned}$$



$$\begin{aligned}
 \omega_1 &= 7.81 \times 10^7 \text{ rps} \\
 \omega_2 &= 12.81 \times 10^7 \text{ rps} \\
 \omega_0 &= 10^8 \text{ rps}
 \end{aligned}$$

(4) d) at ω_0 : L & C together act like short circuit



$$V_{out} = V_{in} \cdot \frac{80}{80+20} = 2 \cdot \frac{4}{5} = \frac{8}{5} \text{ V} \quad (\text{voltage divider})$$

$$V_{out}(t) = \frac{8}{5} \cos(\omega_0 t) \text{ V}$$

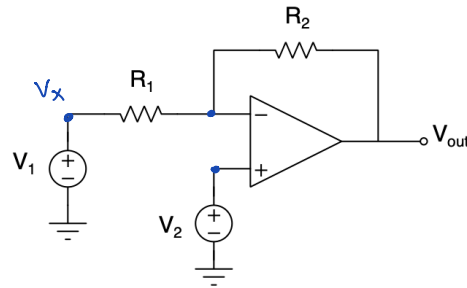
(7) e) at ω_1 :

$$Z_{eq}(\omega_1) = 100 - j100 \, \Omega \quad (\text{see part (c)})$$

$$V_{out} = V_{in} \cdot \frac{R_2}{Z_{eq}(\omega_1)} = 2 \cdot \frac{80}{100 - j100} = \frac{160}{100\sqrt{2} e^{-j\pi/4}} = \frac{4\sqrt{2}}{5} e^{j\pi/4}$$

$$V_{out}(t) = \frac{4\sqrt{2}}{5} \cos(\omega_1 t + \pi/4) \text{ V}$$

Question 3 [25 Points]:



For the circuit given above, assume that the OPAMP is not saturated. In the questions below, A is the open-loop voltage gain of the OPAMP.

- (a) If $A = \infty$, find $H_1(\omega) = \frac{V_{out}}{V_1}$ and $H_2(\omega) = \frac{V_{out}}{V_2}$. Express the transfer functions in terms of R_1 and R_2 . **Hint:** Kill $v_2(t)$ when finding $H_1(\omega)$, and kill $v_1(t)$ when finding $H_2(\omega)$.
- (b) If $A \neq \infty$, find $H_1(\omega) = \frac{V_{out}}{V_1}$ and $H_2(\omega) = \frac{V_{out}}{V_2}$. Express the transfer functions in terms of R_1 , R_2 , and A .
- (c) Find $v_{out}(t)$, if $A = \frac{A_{dc}}{1+j\omega/\omega_o}$ with $A_{dc} = 10^4$ and $\omega_o = 100$ rps, $R_1 = 1 \text{ k}\Omega$, $R_2 = 4 \text{ k}\Omega$, $v_1(t) = 3 \text{ V(DC)}$ and $v_2(t) = 4 \cos(10^6 t) \text{ V}$.

(4) a) $H_1(\omega)$: kill $v_2(t) \Rightarrow V_+ = 0$

For $A = \infty$, $V_+ = V_- = 0$

KCL at V_- : $\frac{0 - V_1}{R_1} + \frac{0 - V_{out}}{R_2} = 0 \Rightarrow H_1(\omega) = \frac{V_{out}}{V_1} = \boxed{-\frac{R_2}{R_1}}$

$H_2(\omega)$: kill $v_1(t) \Rightarrow V_x = 0$,

For $A = \infty$, $V_+ = V_- = V_2$

KCL at V_- : $\frac{V_2 - 0}{R_1} + \frac{V_2 - V_{out}}{R_2} = 0 \Rightarrow H_2(\omega) = \frac{V_{out}}{V_2} = \boxed{1 + \frac{R_2}{R_1}}$

(10) b) $H_1(\omega)$: kill $v_2(t) \Rightarrow V_+ = 0$

For $A \neq \infty$, $V_+ \neq V_-$, $V_{out} = A(V_+ - V_-)$

KCL at V_- : $\frac{V_- - V_1}{R_1} + \frac{V_- - V_{out}}{R_2} = 0 \Rightarrow V_- = \frac{R_2 V_1 + R_1 V_{out}}{R_1 + R_2}$

$V_{out} = A(V_+ - V_-) = A \left(0 - \frac{R_2 V_1 + R_1 V_{out}}{R_1 + R_2} \right)$

$(R_1 + R_2)V_{out} = -AR_2 V_1 - AR_1 V_{out}$

$H_1(\omega) = \frac{V_{out}}{V_1} = \boxed{-\frac{AR_2}{R_1 + R_2 + AR_1}}$

(Note :
 $A \rightarrow \infty$, $H_1(\omega) \rightarrow -\frac{R_2}{R_1}$)

$$\underline{H_2(\omega)} : \text{kill } v_1(t) \Rightarrow V_x = 0, \quad v_+ = v_2$$

$$\underline{\text{KCL at } V_-} : \frac{V_-}{R_1} + \frac{V_- - V_{out}}{R_2} = 0 \Rightarrow V_- = \frac{R_1 \cdot V_{out}}{R_1 + R_2}$$

$$V_{out} = A(v_+ - v_-) = A \left(v_2 - \frac{R_1 \cdot V_{out}}{R_1 + R_2} \right)$$

$$(R_1 + R_2)V_{out} = A(R_1 + R_2)v_2 - A \cdot R_1 \cdot V_{out}$$

$$H_2(\omega) = \frac{V_{out}}{V_2} = \boxed{\frac{A(R_1 + R_2)}{R_1 + R_2 + AR_1}}$$

$$\left(\begin{array}{l} \text{Note:} \\ A \rightarrow \infty, H_2(\omega) \rightarrow 1 + \frac{R_2}{R_1} \end{array} \right)$$

(11) c) use superposition.

* $V_1 = 3 \text{ V}$, with $\omega = 0$

$$A(0) = A_{dc} = 10^4$$

$$H_1(0) = -\frac{10^4 \cdot 4 \cdot 10^3}{5 \cdot 10^3 + 10^4 \cdot 10^3} = -\frac{4 \cdot 10^4}{5 + 10^4} \approx -\frac{4 \cdot 10^4}{10^4} = -4$$

$$V_{out,1} = V_1 \cdot H_1(0) = -12$$

$$V_{out,1}(t) = \boxed{-12 \text{ V (DC)}}$$

* $V_2 = 4 \text{ V}$, with $\omega = 10^6 \text{ rps}$

$$A(10^6) = \frac{10^4}{1 + j10^6/10^2} = \frac{10^4}{1 + j10^4} \approx \frac{10^4}{j10^4} = -j$$

$$H_2(10^6) = \frac{-j(5 \cdot 10^3)}{5 \cdot 10^3 - j \cdot 10^3} = \frac{-j5}{5-j} \cdot \frac{(j)}{(j)} = \frac{5}{1+5j}$$

$$= \frac{5}{\sqrt{26}} e^{j \tan^{-1}(5)} = \frac{5}{\sqrt{26}} e^{-j \tan^{-1}(5)}$$

$$V_{out,2} = V_2 \cdot H_2(10^6) = \frac{20}{\sqrt{26}} e^{-j \tan^{-1}(5)}$$

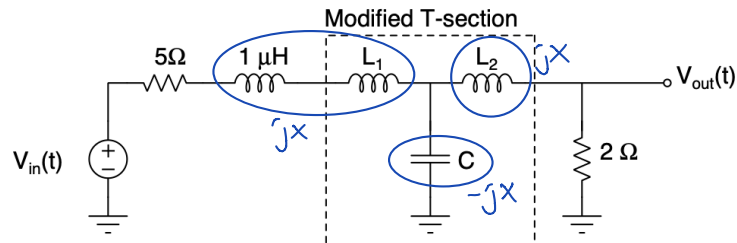
$$\boxed{V_{out,2}(t) = \frac{20}{\sqrt{26}} \cos(10^6 t - \tan^{-1}(5)) \text{ V}}$$

* Finally,

$$V_{out}(t) = V_{out,1}(t) + V_{out,2}(t)$$

$$= \boxed{-12 + \frac{20}{\sqrt{26}} \cos(10^6 t - \tan^{-1}(5)) \text{ V}}$$

Question 4 [20 Points]:



In the circuit above, the modified T-section in the dashed box enables impedance matching between the source and the 2Ω load to maximize power transfer at $\omega_0 = 10^6$ rps.

(a) Find the component values L_1 , L_2 , C for the modified T-section. **Hint:** First, ignore $1\mu H$ inductor and then consider its effect on the T-section.

(b) Find $v_{out}(t)$ if $v_{in}(t) = 3 \cos(\omega_0 t)$ Volts.

(10) a) $1\mu H + L_1$ together forms jX component of T-section.

$$5 = \frac{X^2}{2} \Rightarrow X = \sqrt{10}$$

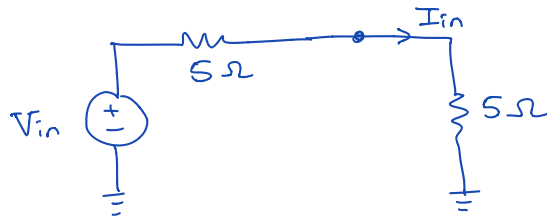
$$jX = j\omega L_2 \Rightarrow L_2 = \frac{X}{\omega} = \sqrt{10} \cdot 10^{-6} = \boxed{3.16 \mu H}$$

$$jX = j\omega(1 \cdot 10^{-6} + L_1) \Rightarrow L_1 = \frac{X}{\omega} - 10^{-6} = \boxed{2.16 \mu H}$$

$$-jX = \frac{1}{j\omega C} \Rightarrow C = \frac{1}{\omega X} = \frac{1}{\sqrt{10} \cdot 10^6} = \boxed{316 \text{ nF}}$$

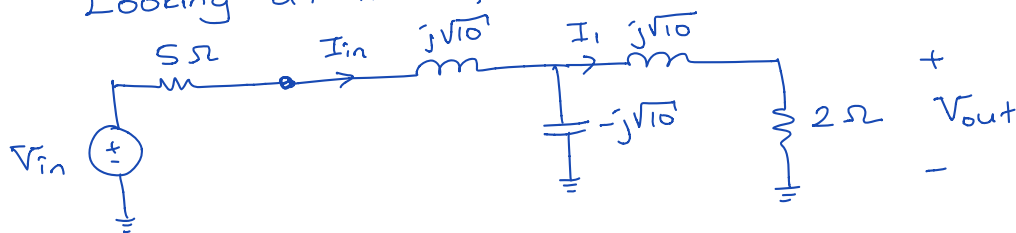
(10) b) $V_{in} = 3V$

Circuit seen by the source:



$$I_{in} = \frac{V_{in}}{10} = \frac{3}{10} \text{ A}$$

Looking at the full circuit

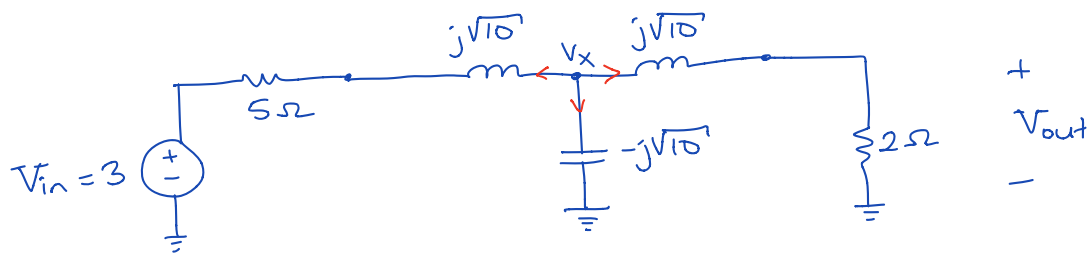


$$I_1 = I_{in} \cdot \frac{-j\sqrt{10}}{-j\sqrt{10} + j\sqrt{10} + 2} = \frac{3}{10} \cdot \frac{(-j\sqrt{10})}{2} = -j\frac{3}{2\sqrt{10}} \text{ A} \quad \left(\text{current divider} \right)$$

$$V_{out} = 2 \cdot I_1 = -j\frac{3}{\sqrt{10}} = \frac{3}{\sqrt{10}} e^{-j\pi/2} \text{ V}$$

$$V_{out}(t) = \frac{3}{\sqrt{10}} \cos(\omega_0 t - \pi/2) \text{ V} = \boxed{0.949 \sin(\omega_0 t) \text{ V}}$$

Alternative solution : Node analysis in phasor domain



KCL at V_x : $\frac{V_x - V_{in}}{5 + j\sqrt{10}} + \frac{V_x}{-j\sqrt{10}} + \frac{V_x}{2 + j\sqrt{10}} = 0$) using calculator

$$V_x = 1.5 - j0.95 = 1.775 e^{-j56}$$

$$V_{out} = V_x \cdot \frac{2}{2 + j\sqrt{10}} = \frac{2 \times 1.775 e^{-j0.56}}{3.74 e^{j1.01}} = 0.949 e^{-j1.57} \text{ V}$$

$V_{out}(t) = 0.949 \cos(\omega_0 t - 1.57) \text{ V}$

 (same result)